Design Note for the Cypress CY7C68000

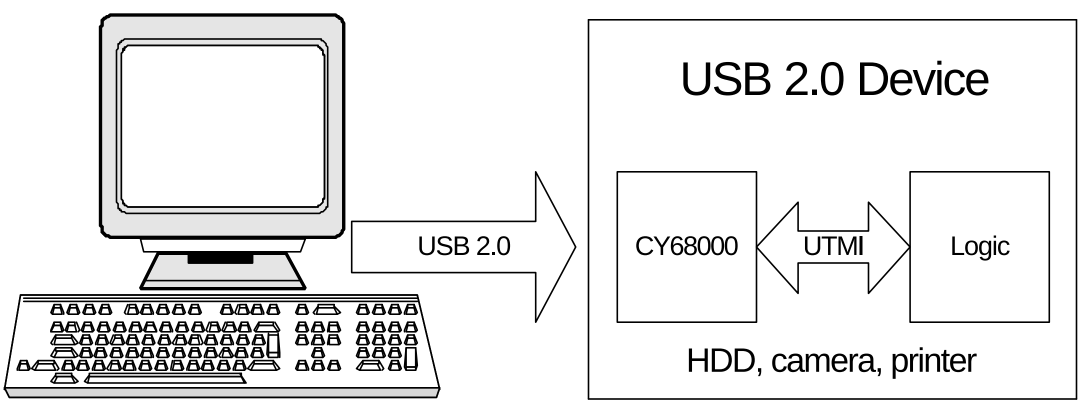
USB 2.0 UTMI PHY

Introduction

The Universal Serial Bus 2.0 Specification (USB 2.0) is an industry standard serial interface between a computer and its peripherals. UTMI is an industry standard USB Transceiver Macrocell Interface between a USB transceiver and another logic device. This design note describes how to design a USB 2.0 device using the Cypress CY7C68000 UTMI Phy chip. This document provides a basic block diagram and schematics that show the typical configuration. Detailed instructions are provided in the CY3683 Development Kit showing design and PCB layout guidelines.

This design note assumes that the reader is familiar with USB 2.0 and UTMI specifications. The USB 2.0 and UTMI specification can be found at the USB Implementers Forum web site[1]. The CY7C68000 data sheet is available from Cypress' website[2].

USB 2.0 UTMI Phy Block Diagram



[1] See http://www.usb.org.
[2] See http://www.cypress.com.

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CY7C68000 Dev Kit Schematic

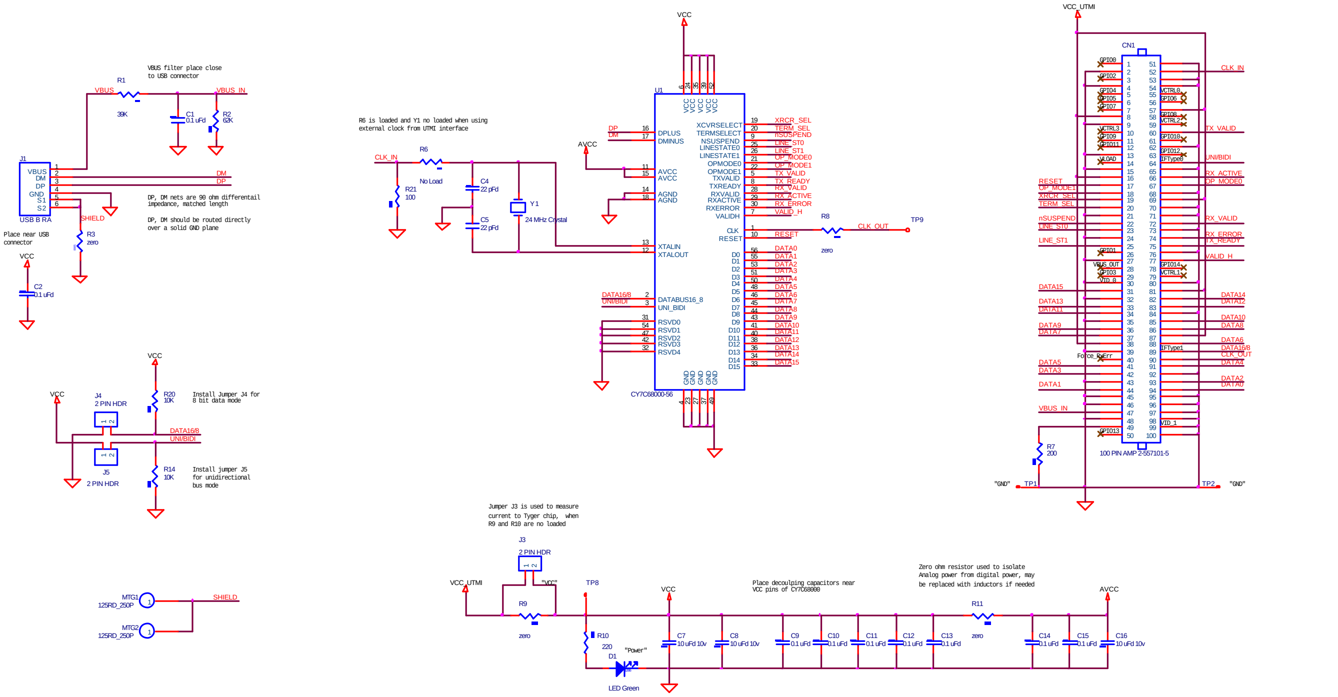

Figure 1: Typical Application of the CY7C68300

See the CY7C68000 Data Sheet for a detailed pin-out description. Reference design schematics are available on the CDROM in the CY3683 Development Kit.

CY7C68000 Design Rules and Layout Guidelines

USB Signals

USB 2.0 signals can transfer data at a rate of 480 Mbits/s on two complementary signals.
The following are required for all USB 2.0 devices:

1. DP and DM trace width, spacing, and PCB layer thickness must be chosen to provide differential impedance of 90 ohms. USB-IF requires DP and DM to have a differential impedance of 90 ohms ± 10%.
2. The maximum trace length for the DP and DM signals is 75 mm. (Recommend 20 to 30 mm).
3. The DP and DM traces must be very close to the same length. The maximum variation in length must be less than 2.5 mm.
4. The PCB manufacturer typically determines DP and DM trace spacing to match the required impedance.
5. Keep all other traces away from DP and DM (6.5 mm min).
6. No vias on DP or DM traces.
7. No right angle bend on DP and DM traces (use 45º angles).

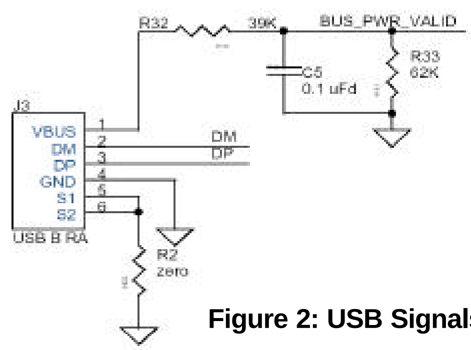

Figure 2: USB Signals

8. A solid ground plane must be placed under the DP and DM traces (no split planes). DP and DM must not be routed over a split power plane.

Power and Capacitance

The CY7C68000 uses an estimated 140mA (max) at 3.3V during normal operation in high-speed (HS) mode. V_{CC} for the CY7C68000 requires a ±10% power supply with clean analog power.

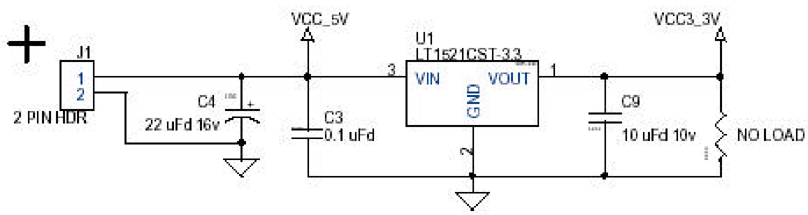

Figure 3: Typical Voltage Regulator Configuration

To adequately power the CY7C68000, a design must include:

1. A 3.3V power supply must supply a minimum of 200mA with ±10% tolerance.
2. The 3.3V power supply must be able to regulate to a load current of 100µA.
3. Use sufficient decoupling capacitors for each active device (the CY7C68000 requires a minimum of one 0.1µF ceramic capacitor for each V_{CC} pin).
4. Some designs may need to isolate the Digital VCC supply from the Analog VCC supply to meet signal integrity requirements. The minimum required bulk storage capacitance for both Digital and Analog V_{CC} is 10µF with a low ESR.
5. If power supply isolation is required Analog V_{CC} must be physically isolated on the PCB from the Digital V_{CC}. Isolating the analog power from digital power with two 1µH inductors is recommended.
6. The Analog Vcc plane must not be broken with traces.

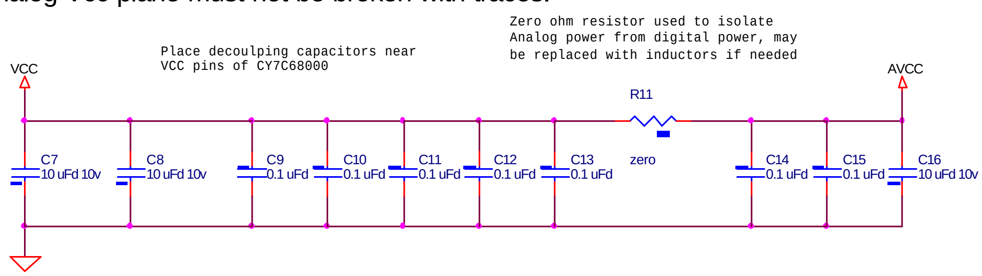

Figure 4: Decoupling Capacitance and Analog Power Isolation

UTMI Signals

1. The UTMI Clock signal should be kept short and routed directly to the corresponding logic device.
2. Ideally the USB 2.0 transceiver should be placed close to corresponding logic device.
3. The UTMI Data traces should be isolated from the UTMI clock and Control signals.

Crystal

1. Crystal requirement is 24 MHz ± 100 ppm, no duty cycle requirements.

2. Load capacitance must be 20pF
3. Oscillation mode must be fundamental
4. The crystal must be placed close to XI and XO pins.
5. The 22pF capacitors must be close to crystal.

PCB Stack-up

1. A typical 4-layer PCB
   Signal (top include DP/DM)
   GND
   V_{CC}
   Signal (plus Analog Vcc)

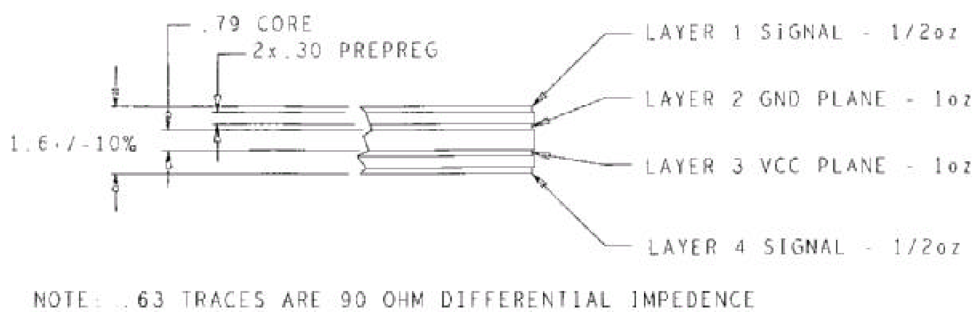

Figure 5: 4-Layer PCB Stack-up

Note: D+ and D- traces are 16mils wide with a 6 mil spacing when using this PCB stack-up.

Document Revision History

Revision #	Date	Comments
1.0	7/16/2002	Initial revision